

```
In [12]: import numpy as np # linear algebra
import pandas as pd # data processing
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [13]: df = pd.read_csv('Social_Network_Ads.csv')
```

```
In [14]: df.head()
```

```
Out[14]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
In [15]: df.describe() # shows the statistics part of numerical data
```

```
Out[15]:
```

	User ID	Age	EstimatedSalary	Purchased
count	4.000000e+02	400.000000	400.000000	400.000000
mean	1.569154e+07	37.655000	69742.500000	0.357500
std	7.165832e+04	10.482877	34096.960282	0.479864
min	1.556669e+07	18.000000	15000.000000	0.000000
25%	1.562676e+07	29.750000	43000.000000	0.000000
50%	1.569434e+07	37.000000	70000.000000	0.000000
75%	1.575036e+07	46.000000	88000.000000	1.000000
max	1.581524e+07	60.000000	150000.000000	1.000000

```
In [ ]:
```

```
In [16]: X=df[["Age","EstimatedSalary"]]
```

```
In [17]: from sklearn.preprocessing import StandardScaler
```

```
In [18]: sc=StandardScaler()
```

```
In [19]: X_scale=sc.fit_transform(X) # fit means Learn transform means convert
```

```
In [20]: X_scale
```

```
Out[20]: array([[ -1.78179743,  -1.49004624],
 [ -0.25358736,  -1.46068138],
 [ -1.11320552,  -0.78528968],
 [ -1.01769239,  -0.37418169],
 [ -1.78179743,   0.18375059],
 [ -1.01769239,  -0.34481683],
 [ -1.01769239,   0.41866944],
 [ -0.54012675,   2.35674998],
 [ -1.20871865,  -1.07893824],
 [ -0.25358736,  -0.13926283],
 [ -1.11320552,   0.30121002],
 [ -1.11320552,  -0.52100597],
 [ -1.6862843 ,   0.47739916],
 [ -0.54012675,  -1.51941109],
 [ -1.87731056,   0.35993973],
 [ -0.82666613,   0.30121002],
 [  0.89257019,  -1.3138571 ],
 [  0.70154394,  -1.28449224],
 [  0.79705706,  -1.22576253],
 [  0.98808332,  -1.19639767],
 [  0.70154394,  -1.40195167],
 [  0.89257019,  -0.60910054],
 [  0.98808332,  -0.84401939],
 [  0.70154394,  -1.40195167],
 [  0.79705706,  -1.37258681],
 [  0.89257019,  -1.46068138],
 [  1.08359645,  -1.22576253],
 [  0.89257019,  -1.16703281],
 [ -0.82666613,  -0.78528968],
 [ -0.63563988,  -1.51941109],
 [ -0.63563988,   0.12502088],
 [ -1.01769239,   1.97500684],
 [ -1.59077117,  -1.5781408 ],
 [ -0.92217926,  -0.75592482],
 [ -1.01769239,   0.59485858],
 [ -0.25358736,  -1.25512738],
 [ -0.44461362,  -1.22576253],
 [ -0.73115301,  -0.60910054],
 [ -1.11320552,   0.06629116],
 [ -1.01769239,  -1.13766796],
 [ -1.01769239,  -1.54877595],
 [ -0.44461362,  -0.55037082],
 [ -0.25358736,   1.123426  ],
 [ -0.73115301,  -1.60750566],
 [ -0.92217926,   0.41866944],
```

```
[-1.39974491, -1.46068138],  
[-1.20871865, 0.27184516],  
[-1.01769239, -0.46227625],  
[-0.73115301, 1.91627713],  
[-0.63563988, 0.56549373],  
[-1.30423178, -1.1083031 ],  
[-1.87731056, -0.75592482],  
[-0.82666613, 0.38930459],  
[-0.25358736, -1.37258681],  
[-1.01769239, -0.34481683],  
[-1.30423178, -0.4329114 ],  
[-1.39974491, -0.63846539],  
[-0.92217926, 0.27184516],  
[-1.49525804, -1.51941109],  
[-0.54012675, 1.38770971],  
[-1.01769239, -1.46068138],  
[-1.20871865, 0.50676401],  
[-1.39974491, -0.10989798],  
[-0.54012675, 1.47580428],  
[ 2.03872775, 0.38930459],  
[-1.30423178, -0.34481683],  
[-1.30423178, -1.49004624],  
[-1.39974491, 0.35993973],  
[-1.49525804, -0.19799255],  
[-0.63563988, -0.05116826],  
[-1.20871865, 0.30121002],  
[-1.30423178, -1.25512738],  
[-1.6862843 , -1.37258681],  
[-0.44461362, 1.27025028],  
[-0.54012675, -1.51941109],  
[-0.34910049, 1.24088543],  
[-1.87731056, -0.52100597],  
[-1.49525804, -1.25512738],  
[-0.92217926, 0.50676401],  
[-1.11320552, -1.54877595],  
[-0.73115301, 0.30121002],  
[ 0.12846516, -0.81465453],  
[-1.6862843 , -0.60910054],  
[-0.25358736, 0.53612887],  
[-0.73115301, -0.2273574 ],  
[-0.63563988, 1.41707457],  
[-1.30423178, -0.4329114 ],  
[-0.92217926, 0.4480343 ],  
[-1.11320552, 0.33057487],  
[-0.25358736, -0.57973568],
```

```
[-1.49525804, 0.33057487],  
[-0.73115301, 1.35834485],  
[-1.11320552, -1.60750566],  
[-0.82666613, -1.22576253],  
[-0.82666613, 0.38930459],  
[-0.25358736, -0.75592482],  
[-0.25358736, -1.3138571 ],  
[-0.92217926, 1.56389885],  
[-0.25358736, 0.09565602],  
[-0.92217926, -0.96147882],  
[-1.01769239, 0.53612887],  
[-0.92217926, -0.31545197],  
[-0.54012675, 0.47739916],  
[-0.44461362, 2.32738512],  
[-1.78179743, -1.43131652],  
[-1.59077117, 0.06629116],  
[-1.11320552, -1.02020853],  
[-1.01769239, 0.56549373],  
[-1.11320552, 0.47739916],  
[ 0.03295203, 0.30121002],  
[ 0.12846516, 0.03692631],  
[-0.0625611 , 0.03692631],  
[ 0.03295203, -0.25672226],  
[-0.0625611 , -0.4329114 ],  
[ 0.41500455, 0.30121002],  
[ 0.22397829, -0.37418169],  
[-0.25358736, 0.15438573],  
[-0.15807423, -0.52100597],  
[ 0.22397829, -0.31545197],  
[ 0.31949142, -0.31545197],  
[-0.15807423, 0.15438573],  
[-0.0625611 , 0.06629116],  
[ 0.22397829, 0.15438573],  
[-0.25358736, -0.49164111],  
[ 0.31949142, -0.55037082],  
[ 0.12846516, -0.25672226],  
[ 0.41500455, -0.13926283],  
[-1.11320552, -1.1083031 ],  
[-0.73115301, -1.54877595],  
[-1.11320552, 0.41866944],  
[-0.63563988, -0.34481683],  
[-0.44461362, -1.13766796],  
[-0.73115301, 0.50676401],  
[-1.59077117, -0.05116826],  
[-0.92217926, -0.4329114 ],
```

```
[-1.39974491, -0.19799255],  
[-1.6862843 , 0.35993973],  
[-0.73115301, 1.09406114],  
[-0.92217926, -0.31545197],  
[-1.78179743, -1.3138571 ],  
[-1.78179743, 0.4480343 ],  
[-1.87731056, -0.05116826],  
[-0.25358736, -0.31545197],  
[-0.73115301, 0.56549373],  
[-0.34910049, -1.3138571 ],  
[-1.30423178, 0.56549373],  
[-1.01769239, 0.77104772],  
[ 0.31949142, -1.16703281],  
[-0.82666613, -0.25672226],  
[-1.6862843 , 0.12502088],  
[-1.11320552, -1.60750566],  
[ 0.31949142, -0.72655996],  
[-0.63563988, 0.18375059],  
[-0.15807423, -0.57973568],  
[ 0.22397829, -0.66783025],  
[-0.63563988, -1.60750566],  
[ 0.79705706, -0.31545197],  
[-0.82666613, 0.15438573],  
[-1.11320552, -1.16703281],  
[-0.54012675, 1.91627713],  
[-0.54012675, 0.88850715],  
[-1.20871865, 0.59485858],  
[-0.0625611 , -1.07893824],  
[-0.25358736, -0.93211396],  
[-0.44461362, -0.02180341],  
[-1.87731056, 0.47739916],  
[-1.49525804, -0.4329114 ],  
[-0.25358736, 0.03692631],  
[-0.82666613, 2.29802026],  
[-0.82666613, -0.66783025],  
[-1.59077117, 0.53612887],  
[-0.34910049, 1.32898 ],  
[-1.11320552, 1.41707457],  
[-0.34910049, -0.78528968],  
[-0.34910049, 0.06629116],  
[-1.39974491, -1.22576253],  
[-0.25358736, -0.66783025],  
[-1.20871865, -1.40195167],  
[-1.30423178, -1.37258681],  
[-0.63563988, -1.04957339],
```

```
[-1.11320552, -1.5781408 ],  
[-0.63563988, 0.03692631],  
[-0.54012675, 1.38770971],  
[-0.44461362, -0.78528968],  
[-0.44461362, -0.28608712],  
[-0.63563988, -0.10989798],  
[-1.6862843 , 0.35993973],  
[-0.44461362, -0.84401939],  
[-0.25358736, 0.06629116],  
[-0.92217926, -1.1083031 ],  
[-1.30423178, 0.41866944],  
[-1.78179743, -1.28449224],  
[-0.82666613, -0.78528968],  
[-1.78179743, 0.00756145],  
[-0.92217926, 0.56549373],  
[-0.34910049, -0.78528968],  
[-0.73115301, 0.27184516],  
[-1.6862843 , -0.99084367],  
[-1.11320552, 0.30121002],  
[-0.25358736, -1.40195167],  
[-0.25358736, -0.9027491 ],  
[ 1.08359645, 0.12502088],  
[ 0.12846516, 1.88691227],  
[ 0.31949142, 0.03692631],  
[ 1.94321462, 0.917872 ],  
[ 0.89257019, -0.66783025],  
[ 1.65667523, 1.76945285],  
[ 1.37013584, 1.29961514],  
[ 0.22397829, 2.12183112],  
[ 0.79705706, -1.40195167],  
[ 0.98808332, 0.77104772],  
[ 1.37013584, 2.35674998],  
[ 2.03872775, -0.81465453],  
[-0.25358736, -0.34481683],  
[ 0.89257019, -0.78528968],  
[ 2.13424088, 1.123426 ],  
[ 1.08359645, -0.13926283],  
[ 0.22397829, 0.2424803 ],  
[ 0.79705706, 0.77104772],  
[ 2.03872775, 2.15119598],  
[ 0.31949142, 0.30121002],  
[-0.25358736, 0.62422344],  
[-0.0625611 , 2.18056084],  
[ 2.13424088, 0.94723686],  
[-0.25358736, -0.28608712],
```

```
[-0.0625611 , -0.49164111],  
[-0.15807423, 1.65199342],  
[ 1.75218836, 1.85754742],  
[ 0.22397829, 0.06629116],  
[ 0.41500455, 0.30121002],  
[-0.25358736, 2.26865541],  
[ 0.12846516, -0.81465453],  
[ 0.22397829, 1.09406114],  
[ 1.08359645, 0.47739916],  
[ 0.03295203, 1.24088543],  
[ 0.79705706, 0.27184516],  
[ 0.22397829, -0.37418169],  
[-0.0625611 , 0.30121002],  
[ 0.79705706, 0.35993973],  
[ 1.46564897, 2.15119598],  
[ 0.41500455, 2.32738512],  
[ 0.03295203, -0.31545197],  
[ 1.17910958, 0.53612887],  
[ 1.75218836, 1.00596657],  
[ 0.31949142, 0.06629116],  
[ 1.27462271, 2.23929055],  
[-0.25358736, -0.57973568],  
[ 1.84770149, 1.53453399],  
[ 0.31949142, -0.52100597],  
[-0.25358736, 0.80041258],  
[ 0.60603081, -0.9027491 ],  
[-0.0625611 , -0.52100597],  
[ 0.98808332, 1.88691227],  
[-0.0625611 , 2.23929055],  
[ 1.17910958, -0.75592482],  
[ 1.37013584, 0.59485858],  
[ 0.31949142, 0.06629116],  
[ 0.22397829, -0.37418169],  
[ 1.94321462, 0.74168287],  
[ 0.70154394, 1.7988177 ],  
[-0.25358736, 0.21311545],  
[-0.15807423, 2.18056084],  
[ 1.65667523, 1.62262856],  
[-0.25358736, 0.06629116],  
[ 0.98808332, 0.59485858],  
[ 0.41500455, 1.123426 ],  
[ 0.22397829, 0.15438573],  
[-0.0625611 , 0.12502088],  
[ 0.89257019, 2.18056084],  
[ 0.22397829, -0.25672226],
```



```
[ 0.51051768, 1.85754742],  
[ 2.03872775, 0.18375059],  
[ 2.13424088, -0.81465453],  
[ 0.12846516, 1.06469629],  
[ 1.84770149, -1.28449224],  
[ 1.84770149, 0.12502088],  
[ 0.03295203, 0.03692631],  
[ 1.08359645, 0.53612887],  
[ 1.37013584, -0.93211396],  
[ 1.17910958, -0.99084367],  
[ 2.03872775, 0.53612887],  
[-0.25358736, -0.25672226],  
[-0.0625611 , 0.00756145],  
[ 1.37013584, -1.43131652],  
[ 0.98808332, 2.09246627],  
[-0.0625611 , 0.68295315],  
[-0.0625611 , -0.2273574 ],  
[ 0.98808332, 2.0043717 ],  
[ 0.31949142, 0.27184516],  
[-0.0625611 , 0.2424803 ],  
[ 0.12846516, 1.88691227],  
[ 1.08359645, 0.56549373],  
[ 1.65667523, -0.9027491 ],  
[-0.0625611 , 0.21311545],  
[-0.25358736, -0.37418169],  
[-0.15807423, -0.19799255],  
[ 0.41500455, 0.09565602],  
[ 0.51051768, 1.24088543],  
[ 0.70154394, 0.27184516],  
[ 0.79705706, 1.38770971],  
[ 1.94321462, -0.93211396],  
[ 0.98808332, 0.12502088],  
[-0.0625611 , 1.97500684],  
[-0.0625611 , 0.27184516],  
[ 0.22397829, -0.28608712],  
[ 0.41500455, -0.46227625],  
[ 1.27462271, 1.88691227],  
[ 0.89257019, 1.27025028],  
[-0.15807423, 1.62262856],  
[ 0.03295203, -0.57973568],  
[ 0.41500455, 0.00756145],  
[ 0.12846516, 0.77104772],  
[ 0.03295203, -0.57973568],  
[ 1.08359645, 2.09246627],  
[ 0.12846516, 0.27184516],
```

```
[ 0.12846516, 0.15438573],  
[ 1.5611621 , 1.00596657],  
[-0.25358736, -0.4329114 ],  
[ 0.70154394, -1.1083031 ],  
[-0.15807423, -0.28608712],  
[ 1.37013584, 2.0043717 ],  
[ 1.46564897, 0.35993973],  
[ 0.31949142, -0.52100597],  
[ 0.98808332, -1.16703281],  
[ 0.98808332, 1.7988177 ],  
[ 0.31949142, -0.28608712],  
[ 0.31949142, 0.06629116],  
[ 0.41500455, 0.15438573],  
[-0.15807423, 1.41707457],  
[ 0.89257019, 1.09406114],  
[ 0.03295203, -0.55037082],  
[ 0.98808332, 1.44643942],  
[ 0.41500455, -0.13926283],  
[ 0.22397829, -0.13926283],  
[ 1.84770149, -0.28608712],  
[-0.15807423, -0.46227625],  
[ 1.94321462, 2.18056084],  
[-0.25358736, 0.27184516],  
[ 0.03295203, -0.4329114 ],  
[ 0.12846516, 1.53453399],  
[ 1.46564897, 1.00596657],  
[-0.25358736, 0.15438573],  
[ 0.03295203, -0.13926283],  
[ 0.89257019, -0.55037082],  
[ 0.89257019, 1.03533143],  
[ 0.31949142, -0.19799255],  
[ 1.46564897, 0.06629116],  
[ 1.5611621 , 1.123426 ],  
[ 0.12846516, 0.21311545],  
[ 0.03295203, -0.25672226],  
[ 0.03295203, 1.27025028],  
[-0.0625611 , 0.15438573],  
[ 0.41500455, 0.59485858],  
[-0.0625611 , -0.37418169],  
[-0.15807423, 0.85914229],  
[ 2.13424088, -1.04957339],  
[ 1.5611621 , 0.00756145],  
[ 0.31949142, 0.06629116],  
[ 0.22397829, 0.03692631],  
[ 0.41500455, -0.46227625],
```

```
[ 0.51051768,  1.74008799],
[ 1.46564897, -1.04957339],
[ 0.89257019, -0.57973568],
[ 0.41500455,  0.27184516],
[ 0.41500455,  1.00596657],
[ 2.03872775, -1.19639767],
[ 1.94321462, -0.66783025],
[ 0.79705706,  0.53612887],
[ 0.03295203,  0.03692631],
[ 1.5611621 , -1.28449224],
[ 2.13424088, -0.69719511],
[ 2.13424088,  0.38930459],
[ 0.12846516,  0.09565602],
[ 2.03872775,  1.76945285],
[-0.0625611 ,  0.30121002],
[ 0.79705706, -1.1083031 ],
[ 0.79705706,  0.12502088],
[ 0.41500455, -0.49164111],
[ 0.31949142,  0.50676401],
[ 1.94321462, -1.37258681],
[ 0.41500455, -0.16862769],
[ 0.98808332, -1.07893824],
[ 0.60603081,  2.03373655],
[ 1.08359645, -1.22576253],
[ 1.84770149, -1.07893824],
[ 1.75218836, -0.28608712],
[ 1.08359645, -0.9027491 ],
[ 0.12846516,  0.03692631],
[ 0.89257019, -1.04957339],
[ 0.98808332, -1.02020853],
[ 0.98808332, -1.07893824],
[ 0.89257019, -1.37258681],
[ 0.70154394, -0.72655996],
[ 2.13424088, -0.81465453],
[ 0.12846516, -0.31545197],
[ 0.79705706, -0.84401939],
[ 1.27462271, -1.37258681],
[ 1.17910958, -1.46068138],
[-0.15807423, -1.07893824],
[ 1.08359645, -0.99084367]]])
```

```
In [21]: X_scale=pd.DataFrame(X_scale)
```

```
In [22]: X_scale.columns=["Age", "EstimatedSalary"]
```

```
In [23]: np.round(X.describe(),2)
```

```
Out[23]:
```

	Age	EstimatedSalary
count	400.00	400.00
mean	37.66	69742.50
std	10.48	34096.96
min	18.00	15000.00
25%	29.75	43000.00
50%	37.00	70000.00
75%	46.00	88000.00
max	60.00	150000.00

```
In [24]: np.round(X_scale.describe(),2)
```

```
Out[24]:
```

	Age	EstimatedSalary
count	400.00	400.00
mean	-0.00	-0.00
std	1.00	1.00
min	-1.88	-1.61
25%	-0.76	-0.79
50%	-0.06	0.01
75%	0.80	0.54
max	2.13	2.36

```
In [ ]:
```